

EDC BLOCK-003 (Derivation v7):

Normalization Candidate Catalog

How to Fix the Constant C in $\kappa_5^2 = C\sigma^{-3/4}$?

EDC Working Document

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Abstract

Derivations v3–v5 established that the 5D gravitational coupling satisfies $\kappa_5^2 = C\sigma^{-3/4}$ with the dimensionless constant C unfixed by brane tension σ alone (No-Go Lemma). This note catalogs 5 candidate normalization principles that could fix C without circularly inserting the observed Newton constant G_N^{obs} . For each candidate, we state the required extra input, assess circularity risk, and evaluate viability. **Outcome:** Catalog complete; one candidate (NC-1: graviton zero-mode normalization) identified as best next attempt. BLOCK-003 remains [OPEN].

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1 Context Recap

From derivations v1–v5:

- **v1:** Standard brane-world reduction yields $G_N = \kappa_5^2/(6\pi L)$ for compact extra dimension of size L .
- **v2:** Testing $L = R_\xi$ (weak scale) is dimensionally consistent but leaves κ_5^2 unknown.
- **v3/v4:** Dimensional analysis uniquely gives $\kappa_5^2 = C\sigma^{-3/4}$, but C cannot be fixed by σ alone (No-Go Lemma: only 2 independent scales available).
- **v5:** Two clean options documented: NP1 (postulate $C = 8\pi$) or NP2 (calibrate to G_N^{obs}).

The question: Can we find a *non-circular* principle that fixes C from EDC-internal or standard physics inputs, without fitting to G_N^{obs} ?

2 Anti-Circularity Definition

Anti-Circularity Criterion

A normalization principle is **circular** if it:

1. Uses G_N^{obs} (or M_{Pl}) as input to fix C .
2. Derives a scale that is itself defined via G_N^{obs} .
3. Tunes a free parameter to match G_N^{obs} predictions.

A principle is **non-circular** if C is determined before any comparison to gravitational observations.

Note: Using G_N^{obs} for *verification* (after C is fixed) is allowed; using it as *input* is not.

3 Candidate Catalog

ID	Name	Fixes C via	Extra Input	Circ.	Status	Notes
NC-1	Zero-mode norm.	Graviton KK reduction	L , BC type	LOW	Viable	Best candidate
NC-2	Induced gravity	DGP-like $R^{(4)}$ term	N , Λ_{UV}	MED	Underdetermined	Species cou unknown
NC-3	RS warp matching	$M_{\text{Pl}}^2 \sim \int e^{-2A}$	$A(\xi)$, k	MED	Missing input	EDC doesn't specify A
NC-4	Topological quant.	Junction/ Z_6 discrete	Topology data	LOW	Speculative	No mechanism yet
NC-5	Bulk Λ_5 match	$\Lambda_5 \sim \kappa_5^2 \rho_P$	Λ_5 value	HIGH	Near-circular	Λ_5 often tuned

Table 1: Normalization candidate catalog. Circularity risk: LOW = no G_N^{obs} dependence; MED = indirect dependence possible; HIGH = likely circular.

4 Candidate Details

4.1 NC-1: Graviton Zero-Mode Normalization

Idea: In Kaluza-Klein reduction, the 4D graviton emerges as the zero-mode of the 5D metric fluctuation. The 4D Planck mass is determined by the normalization integral over the extra dimension.

Setup: Consider the 5D Einstein-Hilbert action

$$S_5 = \frac{1}{2\kappa_5^2} \int d^5 X \sqrt{-g^{(5)}} R^{(5)}.$$

For a factorizable ansatz $ds^2 = g_{\mu\nu}(x)dx^\mu dx^\nu + d\xi^2$ with $\xi \in [0, L]$:

$$S_5 \rightarrow \frac{L}{2\kappa_5^2} \int d^4 x \sqrt{-g^{(4)}} R^{(4)} + \dots$$

Comparing with the 4D action $S_4 = (1/16\pi G_N) \int d^4 x \sqrt{-g} R$:

$$\frac{1}{16\pi G_N} = \frac{L}{2\kappa_5^2} \Rightarrow G_N = \frac{\kappa_5^2}{8\pi L} \quad (1)$$

How it fixes C : If we require the zero-mode to be *canonically normalized* (unit kinetic term), then:

$$\kappa_5^2 = 8\pi G_N L \quad (2)$$

Combined with $\kappa_5^2 = C\sigma^{-3/4}$:

$$C = 8\pi G_N L \cdot \sigma^{3/4} \quad (3)$$

Circularity check:

- If L is specified independently (e.g., $L = R_\xi$), and σ is EDC-internal, then C is fixed *without* using G_N^{obs} as input.
- However, Eq. (3) contains G_N . The question is: can we express G_N in terms of EDC quantities before comparison to observation?

Resolution: Eq. (1) is a *derived relation*, not an input. If we specify L and κ_5^2 (with C from convention), then G_N is a *prediction*. The canonical normalization condition (unit coefficient for $h_{\mu\nu}$ kinetic term) is a standard convention, not a fit.

Required input: Compactification length L (e.g., $L = R_\xi$ [P]).

Status: Viable — best candidate for P18-B attempt.

4.2 NC-2: Induced Gravity (DGP-like)

Idea: Even without a fundamental 5D Einstein term, quantum loops of brane-localized matter induce an effective $R^{(4)}$ term on the brane:

$$S_{\text{induced}} = \frac{M_4^2}{2} \int d^4 x \sqrt{-h} R^{(4)}$$

with $M_4^2 \sim N\Lambda_{\text{UV}}^2/(16\pi^2)$ in 4D.

How it fixes C : The induced 4D Planck mass competes with the bulk-derived contribution. At long distances, the dominant term sets G_N .

Problem: The induced mass depends on:

- N : number of species (unknown in EDC)

- Λ_{UV} : UV cutoff (not specified)
- Regularization scheme (introduces $\mathcal{O}(1)$ factors)

Circularity check: LOW if N and Λ_{UV} are EDC-specified; MED if they require external input or tuning.

Status: Underdetermined — cannot fix C without specifying N and Λ_{UV} .

4.3 NC-3: Randall-Sundrum Warp Matching

Idea: In warped geometries with metric $ds^2 = e^{-2A(\xi)}\eta_{\mu\nu}dx^\mu dx^\nu + d\xi^2$, the 4D Planck mass is:

$$M_{Pl}^2 = \frac{1}{\kappa_5^2} \int_0^L d\xi e^{-2A(\xi)}$$

How it fixes C : If $A(\xi)$ is specified, the integral is computable and relates κ_5^2 to M_{Pl}^2 .

Problem: EDC does not currently specify:

- The warp factor $A(\xi)$
- The AdS curvature scale k (in RS, $A = k|\xi|$)

Circularity check: MED — if k is fit to reproduce correct G_N , this becomes circular.

Status: Missing input — requires EDC to specify bulk geometry or warp factor.

4.4 NC-4: Topological Quantization

Idea: EDC's junction/ Z_6 structure might provide discrete normalization. For example, if the junction action has a topological prefactor (winding number, Euler characteristic), this could quantize C .

How it fixes C : Discrete topology $\Rightarrow C \in \{c_1, c_2, \dots\}$ where c_i are determined by topological data.

Problem: No concrete mechanism exists in current EDC formulation. The Z_6 sector describes particle content, not gravitational normalization.

Circularity check: LOW (topology is internal).

Status: Speculative — attractive in principle but no derivation pathway available.

4.5 NC-5: Bulk Cosmological Constant Matching

Idea: The bulk cosmological constant Λ_5 is related to κ_5^2 via Einstein equations. If Λ_5 is fixed by EDC (e.g., from Plenum energy density), this could constrain κ_5^2 .

How it fixes C : From $\Lambda_5 = -6k^2$ (AdS) and fine-tuning relations, one might extract κ_5^2 .

Problem: In standard RS, the relation is:

$$\sigma = \frac{6k}{\kappa_5^2}$$

which gives $\kappa_5^2 \sigma = 6k$, i.e., $C\sigma^{1/4} = 6k$. This requires knowing k , which is often tuned to hierarchy requirements.

Circularity check: HIGH — Λ_5 or k is typically tuned to reproduce desired G_N or hierarchy.

Status: Near-circular — unlikely to provide non-circular normalization.

ID	Viability	Missing Element	Next Step
NC-1	Viable	L specification	Attempt in P18-B
NC-2	Underdetermined	N, Λ_{UV}	Requires EDC spectrum
NC-3	Missing input	$A(\xi), k$	Requires bulk geometry
NC-4	Speculative	Derivation mechanism	Future research
NC-5	Near-circular	Non-tuned Λ_5	Unlikely path

Table 2: Summary of candidate viability.

5 Summary and Recommendation

Recommendation

Best next attempt: NC-1 (Graviton zero-mode normalization).

Rationale:

- Lowest circularity risk (standard KK procedure, no G_N input).
- Only requires specifying L (can use $L = R_\xi$ as [P]).
- Produces a concrete expression: $C = 8\pi L \sigma^{3/4} G_N$, but with G_N as *output*, not input.
- Clear success/fail criterion: if $G_N^{\text{predicted}}$ matches G_N^{obs} , the model is consistent; if not, it is falsified.

P18-B task: Derive G_N from NC-1 with $L = R_\xi$, compute numerical prediction, compare to observation.

BLOCK-003 Status: Remains [OPEN]. This catalog identifies the path; execution is deferred to P18-B.